

Data Mining Assignment 4 – Classification

Problem Setting: Income Prediction

For this assignment, you will work with the **"income.csv"** dataset. The goal is to train a binary classification model that predicts whether an individual earns more than \$50,000 per year based on demographic and socioeconomic attributes.

Such models are often used in decision-making contexts. Therefore, it is crucial that the models are not only accurate, but also:

- robust against overfitting
- interpretable, so we can understand why predictions are made

Task 1: Building and Evaluating Classification Models

Just like the last assignment, it is important to start with preprocessing the dataset. This may include handling missing values, encoding categorical variables, and normalizing numerical features.

Your first task is to train multiple classification models on the income dataset. Specifically you must experiment with at least two different models of which at least one an ensemble model. You may include additional models if you wish.

For at least two models, explicitly investigate overfitting. You are required to apply at least one technique to reduce overfitting. Clearly report this together with the effect this has.

During your experiments, consider different feature selection techniques and discuss their impact on performance. Ensure careful model evaluation by optimizing hyperparameters where applicable, and designing a proper experimental setup to ensure valid conclusions. Report your findings, which must include at least the following metrics: accuracy, Area Under the Curve (AUC), and precision and recall for both classes.

Task 2: Model Explainability

Select one final model and apply either SHAP or LIME to explain its predictions. Report at least the following:

- Identify the most important features overall.
- Provide explanations for at least two predictions
- Reflect on what the explainability results reveal about your model

Task 3: Applying the Model to New Instances

Select the model from the previous task and apply it to a new dataset, "**income_test.csv**", which contains new persons. Your task is to generate predictions on this new data and provide an estimate of how much they will earn (more or less than \$50.000).

Can you make a reliable estimate of how accurate the model will be and how many people are predicted to earn more than \$50.000? What factors do you base this prediction on? How does your model perform in terms of gender fairness? Discuss briefly how you could improve fairness in your model.

Fill in the **predictions_template.csv** with a prediction for each new person.

Additional Notes on Code and Submission

- Implement your code in any language (Python recommended). Use libraries like:
 - **pandas** for data manipulation
 - **sklearn** to build and evaluate classification models
 - **matplotlib** or **seaborn** for visualizations
 - **shap** or **lime** for explainability
- Include your findings in a concise report of maximum 4 pages.
- **Submit your report as a .pdf with a link to a github repository with your code via Blackboard by 21/05/2026.**
- The PDF file should include your full name like, **Lastname_Firstname-studentNumber.pdf**

Questions: In case you have any questions specific to the assignment, please send an email to nick.wils@uantwerpen.be

Evaluation Criteria

Your submission will be evaluated based on the following grading rubric.

	Less than 7	7 to 10	10 to 13	14 to 17	18 to 20
Writing Style	The report is very confusing; the writing style is below average.	At places the report is not very clear; there is a lack of clear structure.	The text is overall clear although some parts could be improved.	Most of the text is easy to follow, findings are explained in a clear way.	The text is very clearly and concisely written, illustrative examples and figures are not overused, but added where needed.
Task 1	Task was not completed	Task was only completed in a minimal way. Incorrect or insufficient argumentation is given for the choices that were made. The result analysis is not clear.	Task was completed. Some of the explanations behind the methodology/results were okay, though some aspects are missing.	Task was completed and the motivation behind the methodology as well as the result analysis are clear. Some interesting points are made.	The motivation behind the methodology and the result analysis are excellent. Most decisions are backed either by numerical analyses, scientific papers or convincing arguments.
Task 2	Task was not completed	Task was only completed in a minimal way. Incorrect or insufficient argumentation is given for the choices that were made. The result analysis is not clear.	Task was completed. Some of the explanations behind the results were okay, though some aspects are missing.	Task was completed and the motivation behind the analysis is clear. Some interesting points are made.	The motivation behind the analysis is excellent.
Task 3	Task was not completed	Task was only completed in a minimal way. Incorrect or insufficient argumentation is given for the performance estimate.	Effort was done to complete the task but some important aspects are missing.	Most of the task execution was done and motivated well. Only minor aspects have been overlooked.	Complete and thorough analysis was done to complete the task.
Code	Code raises many errors.	Code raises some errors or is very unclear.	Code runs but lacks clear structure and readability, only little	Code is readable and sufficiently documented.	Code is very readable and well documented. It is structured in a way that only by (un)commenting

			documentation is given.		single lines, the code for the different tasks can be run.
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